

LIST OF PROBLEMS: REVENUE EQUIVALENCE, ENVELOPE THEOREM, & EXTENSIONS

ECO403/503–INTRODUCTION TO CONTRACT THEORY AND AUCTIONS

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Note: in all exercises, I am assuming that bidders have private values.

1. *War of Attrition* Consider a two-bidder *war of attrition* in which the bidder with the highest bid wins the object but both bidders pay the losing bid. Bidder's values are independently and identically distributed according to F .
 - (a) Use your preferred method to derive a symmetric, monotonic, equilibrium bidding strategy in the war of attrition.
 - (b) Compute the symmetric equilibrium bidding strategy and the seller's revenue when the bidder's values are uniformly distributed on $[0, 1]$.
2. *Loser-pay auction* Consider a N -bidder *losers-pay auction* in which the bidder with the highest bid wins the object and pays nothing, while all losing bidders pay their own bids. Bidder's valuations are independently and identically distributed according to F .
 - (a) Use your preferred method to derive a symmetric, monotonic, equilibrium bidding strategy in the losers-pay auction.
 - (b) Compute the symmetric equilibrium bidding strategy for the case when the bidders' values are distributed according to $F(v) = 1 - e^{-av}$ over $[0, \infty)$.
3. Consider a N -bidder *third-price auction* in which the bidder with the highest bid wins and pays the *third-highest bid*. Bidder's valuations are independently and identically distributed according to F . Assume that the ratio $\frac{F(v)}{f(v)}$ is increasing in v .
 - (a) Use your preferred method to derive a symmetric, monotonic, equilibrium bidding strategy in the third-price auction. *Hint: Use the revenue equivalence theorem. You will need to differentiate at least two times with respect to v to obtain an expression for the equilibrium strategy $\beta(v)$.*
 - (b) Is the strategy that you derived actually monotonically increasing? What role does the assumption that $\frac{F(v)}{f(v)}$ is increasing in v play in your answer?
4. Suppose that there are two buyers, and that the private values V_1 and V_2 are independent and uniformly distributed over $[0, 1]$. Suppose that the seller intends to sell the object via a first-price auction, but imposes an *entry fee* of λ . In such an auction, a buyer may decide not to participate; for example, he may decline to participate if his private value is lower than the entry fee. If there is only one buyer submitting a bid, that buyer can win the auction by bidding 0. Find the equilibrium of the FPA with an entry fee and compute the seller's revenue as a function of the entry fee. What is the optimal entry fee if any? Can you do the same for a second-price auction with an entry fee?
5. Suppose there are two bidders. Calculate the expected revenue from a first-price SIPV auction as a function of the reserve price r for the following distributions defined over $v \in [0, 1]$:
 - (a) $F(v) = v$
 - (b) $F(v) = v^2$
 - (c) $F(v) = \sqrt{v}$
6. *Discrete values in a private value auction*. Suppose that there is no uncertainty about bidder 1's value and $V_1 = 2$ always. Bidder's 2 value, V_2 , is equally likely to be 0 or 2.
 - (a) Find equilibrium bidding strategies in the first-price auction. *Hint: Note that since values are discrete, the equilibrium will be in mixed strategies.*
 - (b) Compare the revenues in a first and second price auction.
7. *Risk-averse bidders* There are two bidders with private values which are distributed independently according to the uniform distribution over $[0, 1]$. Both bidders are risk-averse and have utility functions $u(z) = \sqrt{z}$. Find a symmetric equilibrium bidding strategy in a first-price auction.